

PORTING-FROM-LAVA

Porting Lava autonomous-QA solutions → Boba-Base (submodules-driven)

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Maps the Lava deep-research playbook (Lava:docs/autonomous-qa/PORTING-PLAYBOOK.md) onto **Boba-Base’s actual stack** (Go qBitTorrent-go + Python download-proxy/merge_service + plugins + frontend/extension + Jackett), respecting Boba-Base’s **TDD + anti-bluff (CONST-XII)** mandate. Both repos already share submodules/{containers, jackett, challenges, helixqa} + constitution, so the reusable pieces land **in the shared submodule** (consumed by both), and only thin glue stays per-repo.

Applicability triage (honest — not a blind copy)

Lava §	Solution	Applies to Boba-Base?	Why
§1	Containerized Android emulator	✗ no	Boba-Base has no Android; UI is web (frontend/extension) → browser E2E (Playwright-in-container) is the analog, separate work
§2	Android guest-net ndc fix	no	Android-only

Lava §	Solution	Applies to Boba-Base?	Why
§7	Compose UI Challenge + off-main nav	no	Android-only
§0/§4	Egress-path insight + VPN-host egress	YES — primary	Same datacenter-IP tracker block; affects Jackett + merge_service + download-proxy + plugin engines
§3	Configurable outbound proxy	YES	download-proxy/qBitTorrent-go/Jackett tracker calls need a configurable egress
§5	Durable remote execution (systemd-linger)	YES	run_all_challenges.sh/ci.sh/helixqa long runs on remote hosts get reaped by logind
§8	Jackett cookie-login + cookie-harvest	partial — already present	Boba-Base has nnmclub-cookie-*, extract-tracker-cookies.sh, extract-jackett-key.py; ADD the apikey-vs-cookie management distinction + test fake-equivalence
§6/§9	Anti-bluff QA verdict + diagnostics	YES	strengthen run_all_challenges.sh verdicts + the diagnostic gotchas

The four high-value ports (in order)

P1 — Durable remote execution (§5) [submodule: containers; glue: scripts/]

Problem Boba-Base shares: long QA/deploy runs on a remote host die when the SSH session ends. **Root cause:** remote systemd-logind KillUserProcesses reaps detached user processes (tmux/nohup/setsid

all die). **Fix:** `loginctl enable-linger + systemd-run --user --unit=<n> --collect bash runner.sh; poll systemctl --user is-active; sentinel` file for completion. Avoid piping the long command through `tail -N` (buffers until exit) — log per-step artifacts. **Port:** add containers submodule helper `pkg/remoteexec` (or `scripts/lib/durable-run.sh`) shared by both repos; wire `scripts/deploy-remote.sh + run_all_challenges.sh` to use it. **TDD:** a test that launches a 60s sleeper via the helper, drops the SSH session, and asserts it's STILL running after (fails against the old `nohup` approach).

P2 — Egress decision + VPN-host routing (\$0/\$4)

[submodule: containers; glue: scripts/ + Jackett/proxy env]

Problem Boba-Base shares: on a datacenter host, trackers are network-blocked (DNS-fail/TLS-MITM, **not** Cloudflare → FlareSolvrr can't fix). Affects Jackett indexer fetches + `merge_service/download-proxy` + plugin engines. **Diagnosis (port the script):** `curl https://api.ipify.org` (host IP) + `curl -o /dev/null -w %{http_code} https://<tracker>/` direct vs via a VPN-host SOCKS proxy. Different egress IP + 200 via proxy ⇒ confirmed. **Fix:** route outbound through a VPN-connected host (the nezha pattern). SOCKS tunnel `ssh -D 127.0.0.1:1080 -N <vpnhost>` (use `--socks5-hostname` for remote DNS); point Jackett + `download-proxy` + `qBitTorrent-go` at it (P3). For browser-cookie harvest, run the harvester ON the VPN host. **Port:** containers submodule `pkg/egress` (tunnel up/verify) + `scripts/egress-via-vpn.sh` glue; reuse Boba-Base's existing `ensure-macos-tunnel.sh` style. **TDD:** assert the via-proxy egress IP ≠ direct host IP AND a known-blocked tracker returns 200 via proxy.

P3 — Configurable outbound proxy in the services (\$3)

download-proxy (Python): `httpx/requests` honor `HTTP_PROXY/HTTPS_PROXY/ALL_PROXY/NO_PROXY` env natively — add an explicit `BOBA_UPSTREAM_PROXY` config that sets these for tracker-bound clients, with loopback bypass (`NO_PROXY=127.0.0.1,localhost,jackett`). **qBitTorrent-go:** set `http.Transport.Proxy` (socks5 native, remote DNS) from a `BOBA_UPSTREAM_PROXY` env — mirror Lava internal/`httpx/proxy.go`. **Jackett:** has a built-in proxy setting (configure via its API/ServerConfig). **Deploy gotcha (port):** the env must be FORWARDED into the containers (`docker-compose.yml` env / the `boba-ctl` deploy) — Lava's bug was a missing allow-list entry. **Verify on distroless via**

podman inspect, not exec printenv. TDD: a test with a local proxy asserts the service's tracker request traverses it; falsifiability: disable the wiring → test fails.

P4 — Jackett cookie-login hardening (§8) [submodule: jackett integration + helixqa fakes]

Add to Boba-Base's existing cookie infra: Jackett's **management API** (/api/v2.0/indexers, indexer /config) needs a **dashboard session cookie** (empty-password POST /UI/Dashboard) — the apikey only authorizes Torznab /results+/caps. If boba-jackett/extract-jackett-key.py only uses the apikey, ListIndexers/config silently 302→/UI/Login. **Critical for anti-bluff:** the test fake MUST 302-without-cookie like real Jackett (behavioral equivalence) or the gap stays hidden. **TDD:** fake 302s management without the cookie; assert discovery succeeds via the cookie path; falsifiability: remove cookie login → 302 failure.

Submodules-driven principle (per Decoupled Reusable Architecture)

- **containers submodule** is the home for P1 (durable-exec) + P2 (egress/tunnel) — generic, both repos + future projects consume. Contribute upstream first, then bump the pin in Lava AND Boba-Base.
- **jackett integration + helixqa** is the home for P4's cookie-login + the behaviorally-equivalent fake.
- **Per-repo glue only:** P3's env wiring (each service's HTTP client + each repo's compose/deploy).

Execution order (TDD per CONST-XII — each step RED→GREEN with pasted evidence)

1. P1 durable-exec helper in containers + wire run_all_challenges.sh/deploy-remote.sh.
2. P2 egress diagnosis + VPN-host tunnel helper in containers + scripts/egress-via-vpn.sh.
3. P3 BOBA_UPSTREAM_PROXY in download-proxy + qBitTorrent-go + Jackett + compose env-forward.
4. P4 Jackett cookie-login + fake-equivalence test.

5. Re-run `run_all_challenges.sh` from the VPN-routed host → real tracker search→download evidence.

Source playbook (full root-cause detail + Lava code refs):

Lava:docs/autonomous-qa/PORTING-PLAYBOOK.md. Each P-item there has the exact mechanism + the falsifiability rehearsal to replicate under Boba-Base's CONST-XII.